Portfolio Project: GER Leather Tannery

PCB Reverse Engineering

Project Details

Project Type:	PCB Reverse Engineering & Industrial Automation
Client / Industry:	Leather Tannery Automation (Industrial)
Tools / Technologies:	KiCad, Multimeter, AI-Assisted Design, PCB Fabrication
Role:	Co-Founder / Lead Electronics Engineer

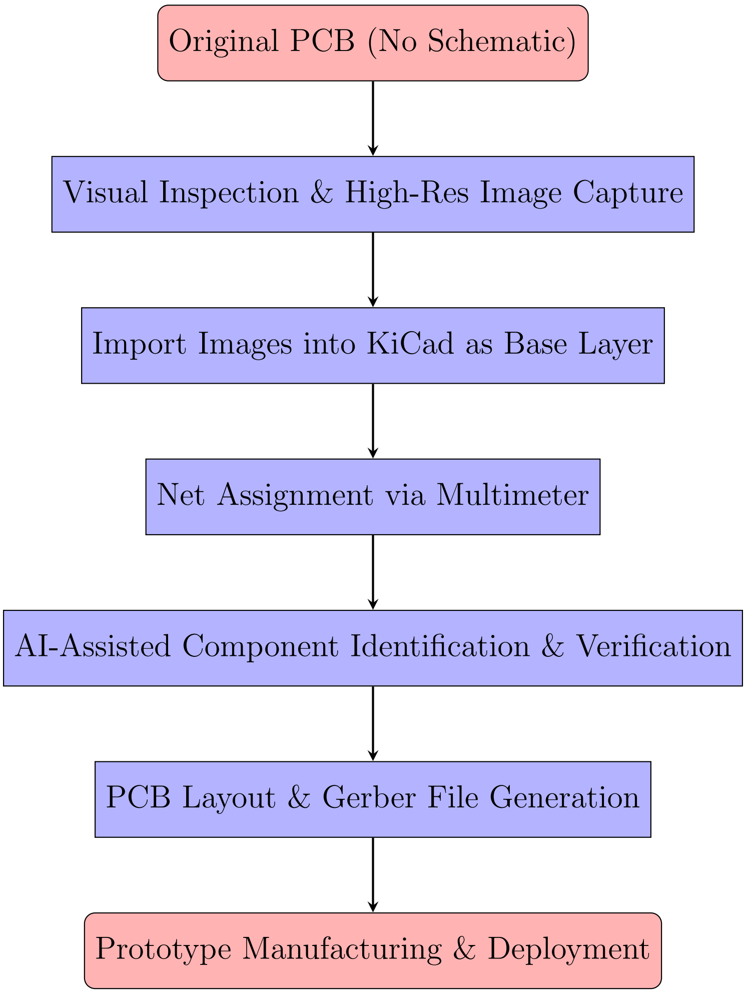
1. Project Overview

We received a PCB from a high-end industrial automation system originally developed by an Italian company (GER). The board had **no schematic** and controlled a critical **automatic spray process** in the leather tannery.

Despite **limited KiCad knowledge**, our confidence, passion, and persistence allowed us to replicate it.

Mission: Reproduce the PCB on paper and in KiCad, fast prototyping, production-ready — *without studying every detail of the original schematic.*

2. Process Flow



3. Challenges & Solutions

Challenge	**Our Approach**
No schematic, complex industrial PCB	Imported PCB images into KiCad, traced visually, measured nets with multimeter, bypassed detailed schematic preparation
Limited KiCad experience	Iterative learning, AI guidance, critical thinking, persistence
Dense routing & mixed signals	Focused on functional replication, fast prototyping, avoided deep analysis
Risk of errors in power/logic circuits	Careful verification, repeated checks, AI-assisted prediction
Tight deadline & high complexity	Discipline, persistence, never gave up — Inshallah

4. Outcome & Impact

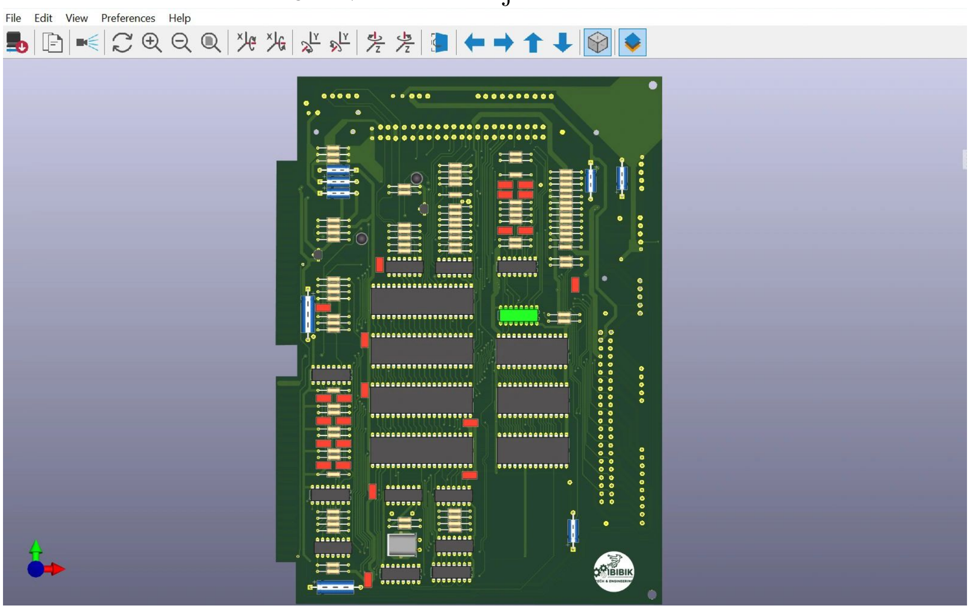
- Successfully replicated high-end industrial PCB
- KiCad layout and Gerber files ready for manufacturing
- Demonstrated problem-solving under constraints: limited knowledge, high complexity, no schematic
- Delivered a reliable, production-ready solution for the client

5. Skills Highlighted

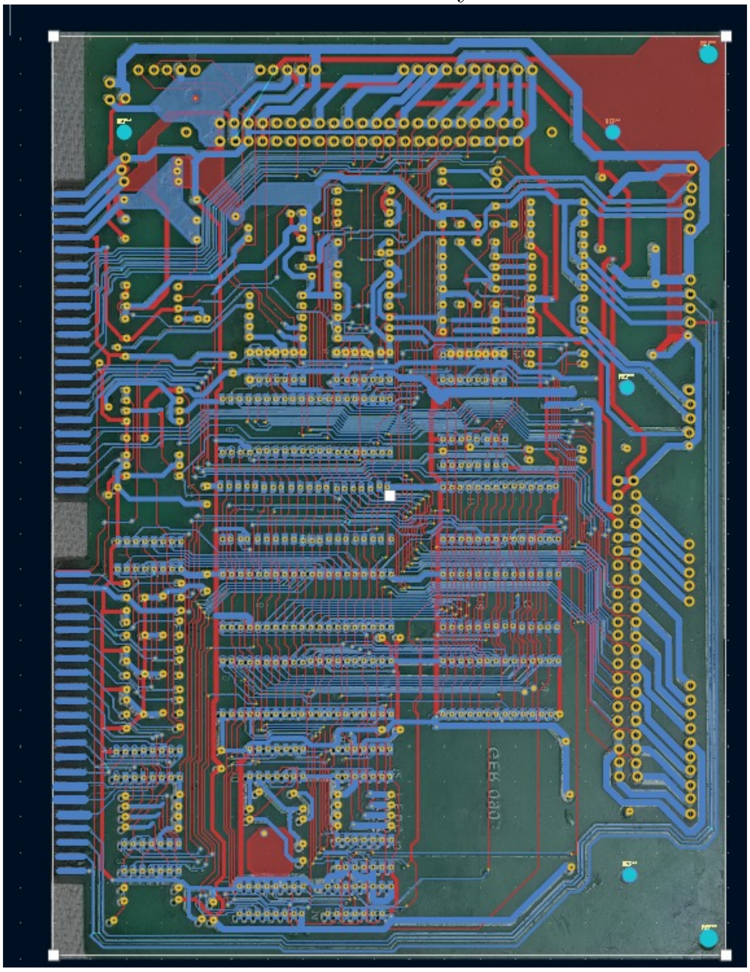
- PCB Reverse Engineering
- KiCad Layout & Gerber Preparation
- Multimeter-Assisted Net Tracing
- AI-Assisted Design Validation
- Fast Prototyping & Production Readiness
- Critical Thinking, Persistence, Problem-Solving

6. Visual Documentation

- 3D Veiw of Project in Kicad



- KiCad schematic & PCB layout screenshot



7. Summary

Even with limited tools and no schematic, we leveraged **critical thinking, AI models, and persistence** to replicate a high-end industrial PCB.

"With determination and modern tools, even complex challenges can be overcome — Inshallah."